



# Equity Pesalink V2 Outbound

Business Service

V 1.0.4

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DEVELOPER API GUIDE DOCUMENT  
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**Revision History**

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# 1.0 Introduction

The Pesalink Outbound Business Service is a client-facing gateway system that facilitates the transfer of funds to external banks and fintechs. This document presents information about the APIs exposed on the system and through which Equity Bank channels such as STK and USSD can initiate name check, credit transfer and transaction status check requests. It also contains information about how the client channels should implement digital signature functionality in order to ensure that the exchange of messages between the channel and the gateway is secure.

## 2.0 API Reference

### 2.1.0 Name Check API

The name check API will return the verified name of the receiver. Based on the response, the customer can decide to proceed with or cancel the credit transfer.

Note: For destination banks and fintechs designated as connecting through the converter as shown in appendix 1 & 2, the name check request will not return a verified name because the implementation is not yet done on the destination. Hence, credit transfer can be initiated even if the name check fails.

*Table 2.1: Name check API endpoint description*

|              |                               |
|--------------|-------------------------------|
| Method       | POST                          |
| Endpoint URL | /v1/verification-request      |
| Headers      | Content-Type:application/json |

#### **Name Check API Request Fields**

*Table 2.2: Name check API request fields*

| Field                   | Description                                                                                                                    | Type   | Mandatory |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|-----------|
| requestId               | Unique identifier for the message                                                                                              | String | Y         |
| channelId               | Unique identifier for the originator channel <sup>1</sup>                                                                      | String | Y         |
| sourceSystemId          | Unique identifier for the source system ID                                                                                     | String | Y         |
| sourceTranType          | Transaction type                                                                                                               | String | Y         |
| senderInstitutionCode   | PIC code of the sender, default value is '0068'                                                                                | String | Y         |
| receiverAccount         | Receiver bank account number or mobile number (for sendToPhone or sendToWallet option). Mobile number format is '254720000000' | String | Y         |
| receiverInstitutionCode | PIC code of the receiver. See appendix 1 for a list of PICs                                                                    | String | Y         |
| signature               | Digital signature generated from the request payload                                                                           | String | Y         |

---

<sup>1</sup> Check section 2.6.0 for default allocated IDs

**Name Check API Response Fields***Table 2.3: Name check API response fields*

| Field                   | Description                                                   | Type   |
|-------------------------|---------------------------------------------------------------|--------|
| requestId               | Unique identifier for the message                             | String |
| endToEndId              | Unique identifier for the transaction                         | String |
| channelId               | Unique identifier for the originator channel                  | String |
| sourceSystemId          | Unique identifier for the source system ID                    | String |
| receiverAccount         | Bank account number of the receiver                           | String |
| receiverInstitutionCode | PIC code of the receiver                                      | String |
| receiverName            | Name of the receiver                                          | String |
| status                  | Status to show whether the message was processed successfully | String |
| statusCode              | Status code ('00' means success, '0x'- means failed)          | String |
| statusDesc              | Status description                                            | String |

**Sample Name Check Request**

```
{
  "requestId": "EQUITY000000000105",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "sourceTranType": "sendToAccount",
  "senderInstitutionCode": "0068",
  "receiverAccount": "0170197176640",
  "receiverInstitutionCode": "0012",
  "signature": "hBJ9FAEzPMaZ37V/iOSTLqxmJfwEBkeC9+cskTmnIf79kxsgbpSyijNAZ3XXxm9kHuN5qPtqJ9tjcbUra9l1Y=="
}
```

**Sample Name Check Response - Success**

```
{
  "requestId": "EQUITY000000000105",
  "endToEndId": "123456789",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "receiverAccount": "0170197176640",
  "receiverInstitutionCode": "0012",
  "receiverName": "John Doe",
  "status": "success",
  "statusCode": "00",
  "statusDesc": "Request Processed Successfully"
}
```

**Sample Name Check Response - Failure**

```
{
  "requestId": "EQUITY000000000105",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "receiverAccount": "1145756190",
  "receiverInstitutionCode": "0001",
  "status": "Failed",
  "statusCode": "01",
  "statusDesc": "Request Failed. Internal Exception Occurred"
}
```

**Name Check Response HTTP Status Codes***Table 2.4: Name check API response HTTP status codes*

| HTTP Status Code | HTTP Status Code Description |
|------------------|------------------------------|
| 200              | Ok                           |
| 202              | Accepted                     |
| 4xx              | Request was not accepted     |



## 2.2.0 Transaction API

The name transaction/credit-transfer API will be used to initiate a transaction.

*Table 2.5: Transaction API endpoint description*

|              |                               |
|--------------|-------------------------------|
| Method       | POST                          |
| Endpoint URL | /api/v1/credit-transfer       |
| Headers      | Content-Type:application/json |

### Transaction API Request Fields

*Table 2.6: Transaction API request fields*

| Field                   | Description                                                                                                                 | Type   | Mandatory |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------|-----------|
| requestId               | Request ID of the request                                                                                                   | String | Y         |
| channelId               | Unique identifier for the originator channel <sup>2</sup>                                                                   | String | Y         |
| sourceSystemId          | Unique identifier for the source system ID                                                                                  | String | Y         |
| callbackURL             | URL that will be used to send a call back request                                                                           | String |           |
| sourceTranType          | Transaction Type. See section 2.6.0                                                                                         | String | Y         |
| sourceAccount           | Bank account number of the sender                                                                                           | String | Y         |
| sourceCountryCode       | Country code of the sender bank account (Use ISO 3166-1 alpha-3 standard)                                                   | String | Y         |
| senderName              | Name of the sender                                                                                                          | String | Y         |
| senderInstitutionCode   | PIC code of the sender, default value is '0068'                                                                             | String | Y         |
| senderPhone             | Mobile phone number of the sender                                                                                           | String | N         |
| receiverAccount         | Receiver bank account number or mobile number (for sendToPhone or sendToWallet option). Mobile number format '254720000000' | String | Y         |
| receiverName            | Name of the receiver. Result of the name check                                                                              | String | N         |
| receiverInstitutionCode | PIC code of the receiver. See appendix 1 for a list of PICs                                                                 | String | Y         |
| receiverCurrency        | Currency code of the receiver bank account (Use ISO 4217 standard)                                                          | String | Y         |

<sup>2</sup> Check section 2.6.0 for default allocated IDs

|               |                                                                                                   |        |   |
|---------------|---------------------------------------------------------------------------------------------------|--------|---|
| amount        | Amount to be sent                                                                                 | String | Y |
| feeAmount     | Amount of transaction fees to be applied as requested by the source channel, default value is '0' | String | Y |
| paymentReason | Reason for payment, e.g: 'School fees'                                                            | String | Y |
| narration     | Narration to be sent with the payment, for example the registration number of student             | String | Y |
| signature     | Digital signature generated from the request payload. See section 3.0                             | String | Y |

### Transaction API Response Fields

Table 2.7: Transaction API response fields

| Field      | Description                                                   | Type   |
|------------|---------------------------------------------------------------|--------|
| requestId  | Request ID of the original transaction request                | String |
| status     | Status to show whether the message was processed successfully | String |
| statusCode | Status code ('00' means success, '0x'- means failed)          | String |
| statusDesc | Status description                                            | String |

### Sample Transaction Request

```
{
  "requestId": "EQUITY000000000222",
  "channelId": "OMNI",
  "sourceSystemId": "OMNI",
  "callbackURL": "http://localhost:8085/v1/callback/channel",
  "sourceTranType": "sendToAccount",
  "sourceAccount": "1001008010940",
  "sourceCountryCode": "54",
  "sourceCurrency": "KES",
  "senderName": "KATEI OLE NASHUU",
  "senderInstitutionCode": "0068",
  "senderPhone": "+254-720000000",
  "receiverAccount": "0170197176640",
  "receiverName": "John Doe",
  "receiverInstitutionCode": "0012",
  "receiverCurrency": "KES",
  "amount": "500",
  "feeAmount": "0",
```

```

    "paymentReason": "Payments",
    "narration": "Pesalink Transfer",

    "signature": "d56Ad+4beRHK/7Eu5AxcbytgAkl4SyWieRTQJvCzev3RDJ0HDYB4pSQOLhKXVhMXDY4EnZp
ZgAEQSWlJ3hYgA8WYpuOkVmlKwOSg5fQKMwo+6ydWEllhzMrsVHJrouze5KIh5aFTR8sXMvGA3bUeOvpzz4T
ZvdzdLH6CYe6Oe2sB+h/2Ukm2lVPbOqjzizwtWETptppVagSwq+oV0rRCNX8hDbFY5W5maiuqrKk3l4fbYlT
KEYASzHlKm1P/tUrlj48WFOhT771OvcdZb+9jz6AQkP8+gPgk/OgKRfujaNgEdM67b3tcZpJTEuy6W+Wwieq
jL/G3udBSs95TwqTAIBw=="
  }

```

### Sample Transaction Response - Success

```

{
  "requestId": "EQUITY000000000105",
  "status": "Success",
  "statusCode": "00",
  "statusDesc": "Request accepted successfully"
}

```

### Sample Transaction Response - Failure

```

{
  "requestId": "EQUITY000000000105",
  "status": "Failed",
  "statusCode": "01",
  "statusDesc": "Request failed"
}

```

### Transaction Response HTTP Status Codes

Table 2.8: Transaction API response HTTP status codes

| HTTP Status Code | HTTP Status Code Description |
|------------------|------------------------------|
| 200              | Ok                           |
| 202              | Accepted                     |
| 4xx              | Request was not accepted     |

### 2.3.0 Payment Status Final Report - Callback API

When Transaction API responds with success and http status code 202, the channel will receive the final transaction status on the callback URL that was supplied with the Transaction API request.

#### Transaction Status Final Report API Response Fields

*Table 2.9: Payment status final report API response fields*

| Field             | Description                                                   | Type   |
|-------------------|---------------------------------------------------------------|--------|
| originalRequestId | The original identifier for the message                       | String |
| endToEndId        | The original identifier for the transaction                   | String |
| status            | Status to show whether the message was processed successfully | String |
| statusCode        | Status code ('00' means success, '0x'- means failed)          | String |
| statusDescription | Status description                                            | String |
| statusMessage     | Status Message                                                | String |

#### Sample Transaction Status Final Report - Success

```
{
  "originalRequestId": "EQ113000000163",
  "endToEndID": "0068006820210705121012c6673fee",
  "status": "Success",
  "statusCode": "00",
  "statusDescription": "Request processed successfully",
  "statusMessage": "Success"
}
```

#### Sample Transaction Status Final Report - Failure

```
{
  "originalRequestId": "EQ00001140",
  "endToEndID": "00680068202106301346416da25d5b",
  "status": "Failed",
  "statusCode": "01",
  "statusDescription": "IPSL responded with reject status.",
  "statusMessage": "Timeout at the Instructed Agent"
}
```

```
}

```

## 2.4.0 Transaction Status Check API

The name transaction status check API will be used to check the status of the transaction initiated during the credit transfer stage based on the information passed during the transaction request.

*Table 2.9: Transaction status check API endpoint description*

|              |                               |
|--------------|-------------------------------|
| Method       | POST                          |
| Endpoint URL | /v1/payment-status-request    |
| Headers      | Content-Type:application/json |

### Transaction Status Check API Request Fields

*Table 3.0: Transaction status check API request fields*

| Field          | Description                                               | Type   | Mandatory |
|----------------|-----------------------------------------------------------|--------|-----------|
| requestId      | Request ID of the request                                 | String | Y         |
| channelId      | Unique identifier for the originator channel <sup>3</sup> | String | Y         |
| sourceSystemId | Unique identifier for the source system ID                | String | Y         |
| signature      | Digital signature generated from the request payload      | String | Y         |

### Transaction Status Check API Response Fields

*Table 3.1: Transaction status check API response fields*

| Field          | Description                                  | Type   |
|----------------|----------------------------------------------|--------|
| requestId      | Request ID of the request                    | String |
| endToEndId     | Unique identifier for the transaction        | String |
| channelId      | Unique identifier for the originator channel | String |
| sourceSystemId | Unique identifier for the source system ID   | String |

<sup>3</sup> Check section 2.6.0 for default allocated IDs

|                         |                                                                                                   |        |
|-------------------------|---------------------------------------------------------------------------------------------------|--------|
| callbackURL             | URL that will be used to send a call back request                                                 | String |
| sourceTranType          | Transaction type                                                                                  | String |
| sourceAccount           | Bank account number of the sender                                                                 | String |
| sourceCountryCode       | Country code of the sender bank account (ISO 3166-1 alpha-3 standard)                             | String |
| sourceCurrency          | Currency code of the sender bank account (ISO 4217 standard)                                      | String |
| senderName              | Name of the sender                                                                                | String |
| senderInstitutionCode   | PIC code of the sender, default value is '0068'                                                   | String |
| senderPhone             | Mobile phone number of the sender                                                                 | String |
| receiverAccount         | Bank account number of the receiver                                                               | String |
| receiverInstitutionCode | PIC code of the receiver                                                                          | String |
| receiverCurrency        | Currency code of the receiver bank account (ISO 4217 standard)                                    | String |
| amount                  | Amount to be sent                                                                                 | String |
| feeAmount               | Amount of transaction fees to be applied as requested by the source channel, default value is '0' | String |
| paymentReason           | Reason for payment, e.g: 'School fees'                                                            | String |
| narration               | Narration captured in the original name check request                                             | String |
| dateAdded               | Date transaction was registered                                                                   | String |
| transactionCount        | Number of transactions                                                                            | String |
| status                  | Status to show whether the message was processed successfully                                     | String |
| statusCode              | Status code ('00' means success, '0x'- means failed)                                              | String |
| statusDesc              | Status description                                                                                | String |
| statusMessage           | Any other information associated with the transaction status                                      | String |

### Sample Transaction Status Check Request

```
{
  "requestId": "EQUITY000000000105",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",

  "signature": "d56Ad+4beRHK/7Eu5AxcbytgAk14SyWieRTQJvCzev3RDJ0HDYB4pSQOLhKXVhMXDY4EnZp
ZgAEQSWlJ3hYgA8WYpuOkVmlKwOSg5fQKMwo+6ydWEl1hzMrsVHJrouze5KIh5aFTR8sXMvGA3bUeOvpzz4T
ZvdzdLH6CYe6Oe2sB+h/2Ukm21VPbOqjzizwtWETptppVagSwq+oV0rRCNX8hDbFY5W5maiuqrKk3l4fbYlT
KEYASzH1Km1P/tUrlj48WFOhT771OvcdZb+9jz6AQkP8+gPgk/OgKRfujaNgeDm67b3tcZpJTEuy6W+Wwieq
jL/G3udBSs95TwqTAIBw=="
}
```

### Sample Transaction Status Check Response - Success

```
{
  "requestId": "EQUITY000000000105",
  "endToEndId": "0068000120210726113234bac05177",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "callbackURL": "http://localhost:8085/v1/callback/channel",
  "sourceTranType": "account",
  "sourceAccount": "0170197176640",
  "sourceCountryCode": "54",
  "sourceCurrency": "KES",
  "senderName": "Jane Doe",
  "senderInstitutionCode": "0068",
  "senderPhone": "+254-720000000",
  "receiverAccount": "1145756190",
  "receiverInstitutionCode": "0001",
  "receiverCurrency": "KES",
  "amount": "500",
  "feeAmount": "0",
  "paymentReason": "Payments",
  "narration": "Pesalinks Transfer",
  "dateAdded": "2021-07-26T11:31:57.275Z",
  "transactionCount": "0",
  "status": "Success",
  "statusCode": "00",
  "statusDesc": "Request processed successfully",
  "statusMessage": "Request was processed successfully by IPS"
}
```

```
}
```

### Sample Transaction Status Check Response - Failure

```
{
  "requestId": "EQUITY000000000105",
  "endToEndId": "0068000120210726113234bac05177",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "callbackURL": "http://localhost:8085/v1/callback/channel",
  "sourceTranType": "account",
  "sourceAccount": "01701971766487",
  "sourceCountryCode": "54",
  "sourceCurrency": "KES",
  "senderName": "Jane Doe",
  "senderInstitutionCode": "0068",
  "senderPhone": "+254-720000000",
  "receiverAccount": "1145756190",
  "receiverInstitutionCode": "0001",
  "receiverCurrency": "KES",
  "amount": "500",
  "feeAmount": "0",
  "paymentReason": "Payments",
  "narration": "Pesalinks Transfer",
  "dateAdded": "2021-07-26T11:31:57.275Z",
  "transactionCount": "0",
  "status": "Failed",
  "statusCode": "01",
  "statusDesc": "Request Failed.",
  "statusMessage": "Request Failed."
}
```



**Transaction Status Check Response HTTP Status Codes***Table 3.2: Transaction status check API response HTTP status codes*

| HTTP Status Code | HTTP Status Code Description |
|------------------|------------------------------|
| 200              | Ok                           |
| 202              | Accepted                     |
| 4xx              | Request was not accepted     |

**2.5.0 MGW Lookup API**

The Mobile Gateway(MGW) Lookup API will be used to fetch a list of banks each with an account number linked to a mobile phone number of a receiver. The bank details contain the name of the receiver as well as the PIC code of the bank. This API is especially useful when a sendToPhone credit transfer request needs to be initiated, as it acts as a substitute for the name-check(account verification request) feature which is not available for the sendToPhone option.

*Table 3.3: MGW Lookup API endpoint description*

|              |                               |
|--------------|-------------------------------|
| Method       | POST                          |
| Endpoint URL | /v1/mgw-lookup                |
| Headers      | Content-Type:application/json |

**MGW Lookup API Request Fields***Table 3.4: MGW Lookup API request fields*

| Field          | Description                                                               | Type   | Mandatory |
|----------------|---------------------------------------------------------------------------|--------|-----------|
| requestId      | Request ID of the request                                                 | String | Y         |
| channelId      | Unique identifier for the originator channel                              | String | Y         |
| sourceSystemId | Unique identifier for the source system ID                                | String | Y         |
| receiverPhone  | Mobile phone number of the receiver; The format should be: '254720000000' | String | Y         |
| signature      | Digital signature generated from the request payload                      | String | Y         |

**MGW Lookup API Response Fields***Table 3.5 MGW Lookup API response fields*

| Field               | Description                                                      | Type            |
|---------------------|------------------------------------------------------------------|-----------------|
| requestId           | Request ID of the request                                        | String          |
| banks               | List of bank objects                                             | List of objects |
| banks[i]            | Bank object                                                      | Object          |
| bank.bankName       | Name of the bank                                                 | String          |
| bank.lookupBankName | Name of the receiver account linked to the provided phone number | String          |
| bank.pic            | PIC code of the receiving bank                                   | String          |
| status              | Status to show whether the message was processed successfully    | String          |
| statusCode          | Status code ('00' means success, '0x' - means failed)            | String          |
| statusDesc          | Status description                                               | String          |
| statusMessage       | Any other information associated with the MGW lookup status      | String          |

**Sample MGW Lookup Request**

```
{
  "requestId": "EQUITY000000000105",
  "channelId": "STK",
  "sourceSystemId": "EQUITEL_STK",
  "receiverPhone": "254722370923",
  "signature": "d56Ad+4beRHK/7Eu5AxcBtYtgAk14SyWieRTQJvCzev3RDJ0HDYB4pSQOLhKXVhMXDY4EnZp
ZgAEQSWlJ3hYgA8WYpuOkVmlKwOSg5fQKMwo+6ydWEllhzMrsVHJrouze5KIh5aFTR8sXMvGA3bUeOvpzz4T
ZvdzdLH6CYe6Oe2sB+h/2Ukm21VPbOqjzizwtWETptppVagSwq+oV0rRCNX8hDbFY5W5maiuqrKk314fbY1T
KEYASzH1Km1P/tUrlj48WFOhT771OvcdZb+9jz6AQkP8+gPgk/OgKRfujaNgeDM67b3tcZpJTEuy6W+Wwieq
jL/G3udBSs95TwqTAIBw=="
}
```

**Sample MGW Lookup Response - Success**

```
{
  "requestId": "EQUITY000000000105",
  "banks": [
    {
      "bankName": "DTB Bank",
      "lookupBankName": "Smith John",
      "pic": "0063"
    },
    {
      "bankName": "NCBA Bank",
      "lookupBankName": "John Smith",
      "pic": "0063"
    }
  ],
  "status": "Success",
  "statusCode": "00",
  "statusDesc": "Fetched successfully",
  "statusMessage": "Fetched successfully"
}
```

**Sample MGW Lookup Response - Failure**

```
{
  "requestId": "EQUITY000000000105",
  "status": "Failed",
  "statusCode": "01",
  "statusDesc": "Request Failed.",
  "statusMessage": "Request Failed."
}
```

*Table 3.6: Transaction status check API response HTTP status codes*

| HTTP Status Code | HTTP Status Code Description               |
|------------------|--------------------------------------------|
| 200              | Ok                                         |
| 4xx              | Request was not accepted or error occurred |

## 2.6.0 Channel and Source System IDs

The following are the channel & source system IDs allocated for the various channels available. Whenever a channel ID is passed in a payload, the correct corresponding source system ID should also be supplied.

*Table 3.7: Channel and source system IDs*

| Channel ID | Source System ID |
|------------|------------------|
| STK        | EQUITEL_STK      |
| APP        | EQUITY_APP       |
| WEB        | EQUITY_WEB       |
| USSD       | EQUITY_USSD      |
| EAZAPP     | EAZZY_APP        |
| EAZNET     | EAZZY_NET        |
| EAZBIZ     | EAZZY_BIZ        |
| EVA        | CHATBOT_EVA      |
| OTC        | BRANCH           |
| OMNI       | OMNI             |

## 2.7.0 Source Transaction Types

The following source transaction types must be supplied for name check and transaction APIs.

Table 3.8: Channel and source system IDs

| Transaction Type | Description                                                              |
|------------------|--------------------------------------------------------------------------|
| sendToAccount    | Use this when sending to a bank account number                           |
| sendToPhone      | Use this when sending to a mobile number linked to a bank account number |
| sendToWallet     | Use this when sending to a Fintech. See appendix 2                       |

## 3.0 Digital Signature

For the purpose of ensuring data integrity and security, every channel will be required to generate a digital certificate and share the same with the maintainer of this API. The certificate will be used to verify the signature contained in each payload.

### 3.0.1 Generating a Cryptographic Key Pair and CSR

One can generate a 2048-bit RSA key using the following Openssl command:

```
openssl req -new -newkey rsa:2048 -nodes -keyout demo.key -out demo.csr
```

Enter the required details as below:

```
Country Name (2 letter code): KE
```

```
State or Province Name (full name): Nairobi
```

```
Locality Name: Nairobi
```

```
Organization Name: Equity
```

```
Organizational Unit Name: pesalink
```

Common Name: equitybankgroup.co.ke

Email Address: info@equitybankgroup.co.ke

Challenge password: demo

## 3.0.2 Generating a Digital Certificate

Once the CSR has been generated, one is required to acquire a valid certificate signed by a recognized CA, but for the purpose of testing, one can generate a self-signed certificate in *.PKCS12* format using the Openssl commands below.

1. Generate a self-signed certificate in *.crt* format:

```
openssl x509 -req -days 365 -in demo.csr -signkey demo.key -sha256 -out  
demo.crt
```

2. Convert the *.crt* certificate to *.p12* format

```
openssl pkcs12 -export -in demo.crt -inkey demo.key -out demo.p12 -name demo
```

3. Share the *.PKCS12* certificate with the maintainer of this API so that it can be configured in the system

## 3.0.3 Generating the Signature

Each request should have a digital signature generated as described below:

1. Generate the clear text:

Concatenate the values in each request field shown below to generate the clear text.  
The clear text value will be hashed.

- a. For Account Verification API, generate the clear text as:

```
clearText = requestId  
          + channelId  
          + sourceSystemId  
          + sourceTranType  
          + senderInstitutionCode  
          + receiverAccount  
          + receiverInstitutionCode
```

- b. For Credit Transfer API, generate the clear text as:  
clearText = requestId  
+ channelId  
+ sourceSystemId  
+ sourceAccount  
+ sourceCurrency  
+ senderName  
+ senderInstitutionCode  
+ receiverAccount  
+ receiverInstitutionCode  
+ receiverCurrency  
+ amount  
+ feeAmount
  - c. For Status Check API, generate the clear text as:  
clearText = requestId
  - d. For MGW lookup API, generate the clear text as:  
clearText = requestId  
+ receiverPhone;
2. Sign the clear text:  
Sign the request using the clear text generated in step 1, the RSA private key generated in section 3.0.2 and by using the SHA-256 hashing algorithm.  
In summary, the signature scheme should be SHA-256 with RSA

### 3.0.4 Sample Code(Java)

The figure below shows an example of how the signature can be generated using Java although the same can be achieved using any programming language

*Fig. 1: Example of how to generate a digital signature using Java*

```
import java.security.PrivateKey;
import java.security.Signature;
import java.util.Base64;

public class SignMsg {

    private static final String UTF_8 = "UTF-8";
    private static final String ALGORITHM = "SHA256WithRSA";

    public String signMessage(String theXsumFields, PrivateKey privatekey) {
        try {
            byte[] messageBytes = theXsumFields.getBytes(UTF_8);
            Signature signatureInstance = Signature.getInstance(ALGORITHM);
            signatureInstance.initSign(privatekey);
            signatureInstance.update(messageBytes);
            byte[] digitalSignature = signatureInstance.sign();
            String signature = Base64.getEncoder().encodeToString(digitalSignature);
            return signature;
        } catch (Exception ex) {
            ex.printStackTrace();
            return null;
        }
    }
}
```



## Appendix 1: List of Participant Identification Codes(PIC) For Banks

| BANK NAME            | PARTICIPANT IDENTIFICATION CODE(PIC) | CONNECTING THROUGH |
|----------------------|--------------------------------------|--------------------|
| KCB                  | 0001                                 | DIRECT             |
| Stanchart            | 0002                                 | CONVERTOR          |
| ABSA                 | 0003                                 | DIRECT             |
| Bank of India        | 0005                                 | DIRECT             |
| Bank of Baroda       | 0006                                 | DIRECT             |
| NCBA                 | 0007                                 | DIRECT             |
| Prime Bank           | 0010                                 | DIRECT             |
| Coop Bank            | 0011                                 | DIRECT             |
| NBK                  | 0012                                 | DIRECT             |
| M-Oriental           | 0014                                 | DIRECT             |
| Citi Bank            | 0016                                 | CONVERTOR          |
| Habib Bank AG Zurich | 0017                                 | DIRECT             |
| Middle East Bank     | 0018                                 | CONVERTOR          |
| Bank of Africa       | 0019                                 | DIRECT             |
| Consolidated         | 0023                                 | DIRECT             |
| Credit Bank          | 0025                                 | DIRECT             |
| Access Bank          | 0026                                 | DIRECT             |
| Stanbic Bank         | 0031                                 | DIRECT             |
| ABC Bank             | 0035                                 | CONVERTOR          |
| Eco Bank             | 0043                                 | DIRECT             |
| SPIRE Bank           | 0049                                 | DIRECT             |
| Paramount            | 0050                                 | DIRECT             |
| Kingdom Bank         | 0051                                 | CONVERTOR          |
| Gt Bank              | 0053                                 | CONVERTOR          |
| Victoria Bank        | 0054                                 | CONVERTOR          |
| Guardian Bank        | 0055                                 | CONVERTOR          |
| I&M Bank             | 0057                                 | DIRECT             |
| Development Bank     | 0059                                 | DIRECT             |
| SBM                  | 0060                                 | DIRECT             |
| Housing finance      | 0061                                 | CONVERTOR          |

|                              |      |           |
|------------------------------|------|-----------|
| DTB                          | 0063 | DIRECT    |
| Mayfair Bank                 | 0065 | DIRECT    |
| Sidian Bank                  | 0066 | CONVERTOR |
| Equity Bank                  | 0068 | DIRECT    |
| Family Bank                  | 0070 | CONVERTOR |
| Gulf African Bank            | 0072 | CONVERTOR |
| First Community Bank         | 0074 | CONVERTOR |
| DIB Bank                     | 0075 | CONVERTOR |
| UBA                          | 0076 | DIRECT    |
| KWFT                         | 0078 | CONVERTOR |
| Faulu Bank                   | 0079 | DIRECT    |
| Post Bank                    | 0099 | DIRECT    |
|                              |      |           |
| IPS (Instant Payment Switch) | 9999 |           |

## Appendix 2: List of Participant Identification Codes(PIC) For Fintechs

| <b>FINTECH NAME</b> | <b>PARTICIPANT IDENTIFICATION CODE(PIC)</b> | <b>CONNECTING THROUGH</b> | <b>TEMPORARY PIC ASSIGNED</b> |
|---------------------|---------------------------------------------|---------------------------|-------------------------------|
| Telkom Kenya        | 0097                                        | DIRECT                    | 0097                          |
| Airtel Kenya        | TO BE CONFIRMED                             | DIRECT                    |                               |
| Eclectics           | 0096                                        | DIRECT                    | 0096                          |
| MFS                 | 0095                                        | IPSL PAYMENT GATEWAY      | 0095                          |
| Kocela              | 0094                                        | IPSL PAYMENT GATEWAY      | 0094                          |
| KCBbulk             | 0093                                        | IPSL PAYMENT GATEWAY      | 0093                          |
| iPay                | 0092                                        | IPSL PAYMENT GATEWAY      | 0092                          |
| Flutterwave         | 0091                                        | IPSL PAYMENT GATEWAY      | 0091                          |
| Cellulant           | 0090                                        | DIRECT                    | 0090                          |
| Stima Sacco         | 0089                                        | IPSL PAYMENT GATEWAY      | 0089                          |
| Interswitch         | 0088                                        | IPSL PAYMENT GATEWAY      | 0088                          |
| Craft Silicon       | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Kenswitch           | TO BE CONFIRMED                             | DIRECT                    |                               |
| Pesapal             | 0087                                        | IPSL PAYMENT GATEWAY      | 0087                          |
| Pesaswap            | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| MyKeekapu           | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Sybyl Kenya Ltd     | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Power               | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Sevi                | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Virtual City        | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| KAPS                | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |
| Pendo Media         | TO BE CONFIRMED                             | TO BE CONFIRMED           |                               |